Should We Use Track Records to Identify Experts?

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Abstract. Huang (2023) argues against the common view that track records are the best way to identify experts. She uses an agent-based coin-flip model to show that, under plausible conditions, track records can fail to help a community identify experts and reduce average group accuracy. I present several modifications of her model to make it more realistic and which show why track records are likely to be helpful in real-life communities.

1 Introduction

Experts are often judged by their track records. If someone is truly an expert, then we would expect them to be able to make more accurate predictions than non-experts and so have a better track record over time. For example, a chess player's ability is tracked over time by their elo rating which is a measure of their track record against other players. Relatedly, in the academic world, researchers are often judged by their track record of publications and citations. The idea is that if someone has a good track record, then we have good reason to trust their testimony and so to defer to them on matters within their area of expertise. This is a common view of how to identify experts and so it has important implications for how we should structure our academic communities and how we should make public policy decisions about who to listen to on important issues.

However, Huang has surprisingly found that track records can fail to identify experts and can even reduce group accuracy under plausible conditions (Huang, 2023). This is a surprising finding since it goes against the common view that track records are the best way to identify experts. If track records can fail to identify experts, then we need to reconsider how we identify experts and how we structure our academic communities.

In what follows, I'll begin by describing Huang's model and explaining her results. I'll then strengthen her findings by showing that they hold across a different kind of expertise than the one she models. Finally, I'll identify three features of real-life academic communities which undermine Huang's findings and vindicate track records: (i) the fact that track records are often *unclear and delayed*, (ii) the fact that track records are often *less trusted early on*, and (iii) the fact that track records are often *based on public evidence*.

2 Huang on Track Records

Huang (2023) argues that track records may plausibly fail to guide us to experts. To do this, she presents an agent-based model she believes to be plausible and in which average accuracy using track records is lower than without using track records. The kind of community she's considering is one of quasi-peers who can all generate and interpret evidence but some interpret it better than others. The idea is that if quasi-peers in an academic community can't reliably identify who the experts are among themselves using track records, then *a fortiori* the general public and/or policymakers can't either. Let's first consider the model to understand it and then consider her results.

2.1 Huang's Model

The basic idea behind Huang's model goes as follows. Agents have opinions about the true bias of a coin with an unknown bias towards landing heads. They each give their own probability that the coin will land heads on the next flip and then make a private flip and observe the outcome. They each then update their own opinions using two factors: Their own opinion of what the bias of the coin is given their private coin flips that they've observed (the evidential component) and the opinions of other agents with the best track records (the social component). The performance of the agents is measured by the average accuracy of their predictions across all flips. Readers who don't want the technical details are encouraged to skip the remainder of this section and go straight to Section 2.2.

To be more precise, Huang's model involves a group of $N = 30$ agents who each make private flips of a common coin with an unknown bias towards heads at each time step. We suppose that the possible biases of the coin are 0, 0.2, 0.4, 0.6, 0.8, or 1. Let H_j represent the hypothesis that the coin has bias j where $j \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$. Agents assign personal probabilities to each of these biases being the true bias of the coin. Each agent starts with an equal probability of $1/6$ for each bias.[1] At each time step t, they use their

[1] That is, each agent starts with the credence function: $\Pr(H_0) = \Pr_0(H_{0.2}) = \Pr(H_{0.4}) = \Pr(H_{0.6}) = \Pr(H_{0.8}) = \Pr(H_1) = 1/6$.

personal probabilities to predict the probability that the next flip of the coin will land heads. For instance, for the first flip, every agent would predict a 0.5 probability of heads.[2] They then each make a private flip of the coin and privately observe the outcome. Their average accuracy across all flips (measured using the Brier score) which is the measure of their track record is updated and then made public.[3] Each agent then updates their personal probabilities using a linear combination of an evidential component $\Pr_t^e(H_j)$ and a social component $\Pr_t^s(H_j)$ (Hegselmann & Krause, 2002):

$$\Pr_t(H_j) = c\Pr_t^e(H_j) + (1-c)\Pr_t^s(H_j) \tag{1}$$

This process then repeats for a given number of time steps.

The evidential component $\Pr_t^e(H_j)$ for each agent is calculated by updating the agent's personal probabilities using Bayes' theorem after each coin flip with some normally distributed noise b depending on how much of an expert they are.[4] The social component $\Pr_t^s(H_j)$ is calculated as the average personal probabilities of the $k = mN$ agents with the

[2] The likelihoods of observing heads on the next toss (H) given each bias are: $\Pr(H|H_0) = 0$, $\Pr(H|H_{0.2}) = 0.2$, $\Pr(H|H_{0.4}) = 0.4$, $\Pr(H|H_{0.6}) = 0.6$, $\Pr(H|H_{0.8}) = 0.8$, and $\Pr(H|H_1) = 1$. $\Pr(H)$, their expected probability of flipping a heads on the next toss, can be calculated using the law of total probability: $\Pr(H) = \Pr(H|H_0)\Pr(H_0) + \Pr(H|H_{0.2})\Pr(H_{0.2}) + \Pr(H|H_{0.4})\Pr(H_{0.4}) + \Pr(H|H_{0.6})\Pr(H_{0.6}) + \Pr(H|H_{0.8})\Pr(H_{0.8}) + \Pr(H|H_1)\Pr(H_1) = (0)(1/6) + (0.2)(1/6) + (0.4)(1/6) + (0.6)(1/6) + (0.8)(1/6) + (1)(1/6) = 1/2$.

[3] The Brier score is calculated as follows:

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N}(f_i - o_i)^2$$

where N is the total number of predictions, f_i is the predicted probability of heads for the i-th flip, and o_i is the actual outcome (1 for heads, 0 for tails). This is essentially just mean-squared error. A lower Brier score indicates better accuracy.

[4] Let's work through an example to see how this updating works. Suppose an agent observes a heads on the first flip. They then update their probailities of each bias using Bayes' theorem:

$$\Pr(H_j|E) = \frac{\Pr(E|H_j) \times \Pr(H_j)}{\Pr(E)}$$

For example, for bias $j = 0.8$:

$$\Pr(H_{0.8}|E) = \frac{\Pr(E|H_{0.8}) \times \Pr(H_{0.8})}{\Pr(E)} = \frac{(0.8) \times (1/6)}{(1/2)} \approx 0.2667$$

The probability that the coin has bias 0.8 given the observed heads has risen from 0.1667 to 0.2667. Next, some normally-distributed noise will be added to each $\Pr(H_j|E)$ with a mean of $\Pr(H_j|E)$ and standard deviation b. Finally, these personal probabilities will be normalized so that they sum to 1.

best track record, where m is a parameter unique to each agent.[5] For instance, if an agent has $m = 0.1$ and there are $N = 30$ agents, then that agent would calculate their social component as the average of the personal probabilities of the $k = 3$ agents with the best track records. These evidential and social components are calculated after each coin flip and then combined in accordance with Equation (1).

We can note that this model attributes three important properties to each agent. First, each agent has their own expertise level (b) which is uniformly-randomly drawn from [0,0.2]. This represents how much noise there is when updating their personal probabilities given their new evidence from their coin flip. An agent's actual evidential component for each bias is drawn from the normal distribution with mean equal to what their new personal probability should be (given probability theory) and standard deviation equal to b. This means that around 99% of the time, their evidential component will be within $\pm 3b$ of the rational personal probability. An agent with $b = 0$ is a perfect Bayesian updater. This means that expertise in this model is understood not as the ability to *generate* better data but the ability to *interpret* the epistemic implications of their perfect data better than others.

Second, each agent has an oppenness level (m) which is uniformly-randomly drawn from [0.05,0.4]. This represents how many of the other agents they consider when calculating their social component. This means that agents select anywhere from the top 5% to the top 40% of agents based on track record to inform their social component. For $N = 30$ this means that agents consider anywhere from the top 2 agents to the top 12 agents when calculating their social component.

Third, each agent has a weighting factor (c) which is uniformly-randomly drawn from [0,1]. This represents how much weight they give to their evidential component versus their social component. An agent with $c = 0$ only considers the opinions of others while an agent with $c = 1$ only considers their own opinions.

[5] Formally:

$$\mathrm{Pr}^s_t(H_j) = \frac{\sum_{i=1}^{k} \mathrm{Pr}_{i,t-1}(H_j)}{k}$$

$\mathrm{Pr}_{i,t-1}(H_j)$ is the credence function of the i-th informant at time $t-1$ and k is the total number of informants for the agent.

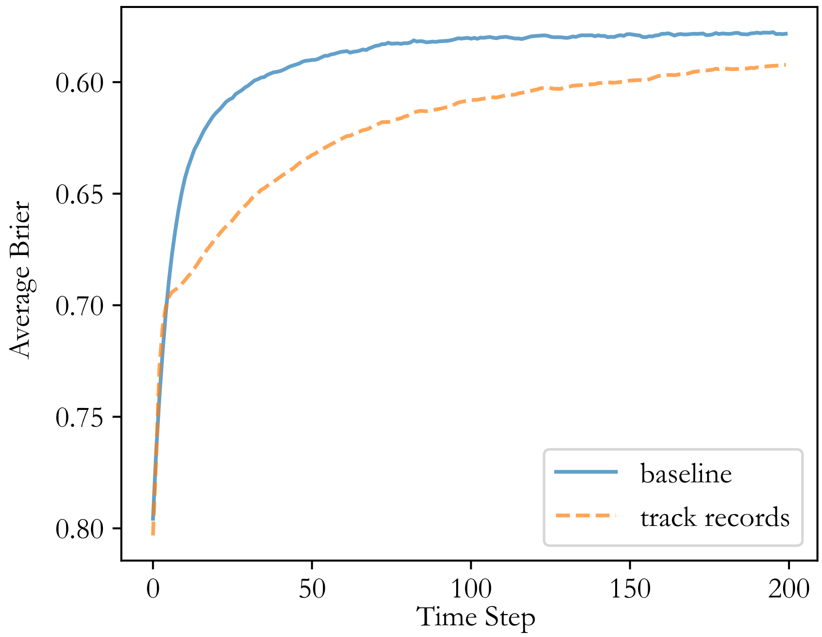

Figure 1: Track Records vs. Baseline

2.2 Huang's Results

Huang ran 600 runs of this model for 100 time steps under two different conditions. First, there was a baseline model where agents only used their own opinions ($c = 1$ for all agents). This shows how effective the agents are working alone with no regard for track records. Second, there was a track records model where agents used both their own opinions and the opinions of those with good track records as described above. I recomputed the results over 200 steps (rather than 100) and averaged over 10,000 runs (rather than 600). The results are shown in Figure 1.

What Huang found was surprising: Track records actually made agents perform worse than the baseline condition where they ignored track records entirely. That is, the average accuracy when agents used track records was lower than when they ignored track records. Huang explained this result as being caused by opinion monopoly operating on sparse data. Early on, some agents accumulate good track records by chance because their data better represented the true bias of the coin regardless of their actual level of expertise. Since such agents have the best track records, the other agents begin to accomodate themselves to these agents' opinions thinking them to be the experts. This reduces epistemic diversity without any actual gain in accuracy since these agents were only lucky rather than skilled. As agents' opinions converge to one another, it also becomes more difficult to discern which agents are actually experts since they make similar predictions. As a result, the entire group performs worse than if they had ignored track records entirely.

Huang considers two possible ways in which to amerliorate this problem. First, she modifies the model so that half of the agents choose their informants randomly rather than based on track records. This is the "less monopoly" model since it reduces the degree of opinion monopoly. Second, she modifies the model so that agents wait until time step 51 before they begin using track records to select informants.[6] The idea is that this allows agents to accumulate large sample size track records which have been influenced only by their expertise and not the opinions of others before people start using the track records. This is the "patience" model since it requires agents to be patient before relying upon track records. Once again I recomputed the results, which are shown in Figure 2. The only significant difference I found with Huang's original results is the somewhat better performance of the less monopoly model, probably due to the larger number of runs that I averaged over.[7]

The less monopoly model provides improvement over the track records model and even surpasses it after around time step 100. The patience model performs identically to the baseline until time step 51, at which point it jumps in accuracy and stays above the baseline for all time steps. This shows just how valuable it is to get large sample size track records uncontaminated by social influence. The accuracy of the model regresses after this time step and approaches the accuracy of the baseline (though always remaining slightly better). Huang doesn't explain this regression in her paper but it's probably due to the difficulty in discerning experts once non-experts can converge their opinions to the true experts.

Nevertheless, Huang argues persuasively that both of these models are unrealistic. The less monopoly model requires *some* agents to *permanently* ignore track records when choosing informants. The patience model requires *all* agents to *temporarily* ignore track records for a sustained period of time. Each of these involves a collective action problem since it requires at least some agents to not pay heed to track records which probably isn't in their best interest.

Huang draws the conclusion that soliciting testimony based on perfectly accessible

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[6] This involves changing the value of c from 1 to its randomly-assigned value at time step 51.

[7] See Huang (2023, Figure 5).

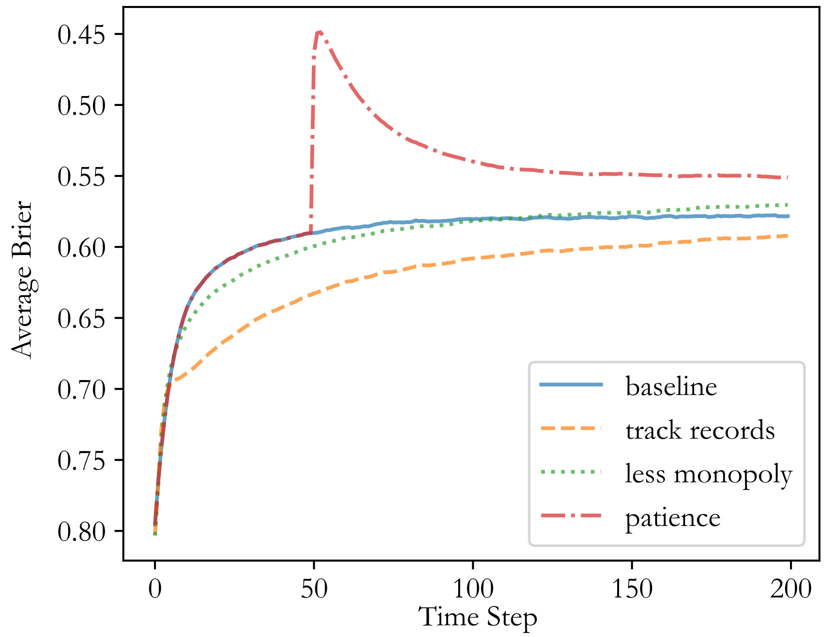

Figure 2: Huang's Models of Track Records

and evaluable track records can fail to guide one to the experts and can actually reduce group accuracy. We face a dilemma: Wait a while for track records to be useful or use them immediately and do worse than not using them at all. In other words, track records aren't useful at the start of inquiry. Huang recognizes at the end of her paper that, "There is a lot more work to be done here, and plenty of other potential solutions to be studied." I'll now take up this task.

3 Expert Priors

I'll now present four possible changes to Huang's model. The purpose of these changes is to investigate the robustness of Huang's findings as we make the model more realistic. The first modification I want to look at considers different kinds of expertise. Recall that the expertise in Huang's model is entirely about how well agents *interpret* evidence. All agents start by assigning equal personal probability to each potential bias of the coin being the true bias but experts update their opinions more accurately than non-experts. Let's call this kind of expertise *interpretational expertise*.

However, there are other kinds of expertise we might be interested in. I'll mention two. First, there's *experimental expertise* which is the ability to generate better data. While some scientists are theoreticians who focus on developing better models to explain the data that we have, others are experimentalists who focus on generating better data. It doesn't seem that experimental expertise is captured in Huang's model since all scientists obtain perfect data from the coin flips. Second, there's *prior expertise* which is the possession of better priors when encountering a new research problem. An expert's experience in related areas may give them a better idea of the plausible solutions to the new problem. If a new pandemic arises, biologists and public health experts would likely have better priors about the characteristics of the virus than laypeople given their experience with other viruses. Prior expertise also isn't captured in Huang's model since all agents start in the same state of complete ignorance.

Given that agents generate binary data in Huang's model (heads or tails), it isn't straightforward to model experimental expertise without changing the update mechanism. Instead, I'll just focus on prior expertise. We can model prior expertise by having agents update their priors with some warm up tosses before the actual simulation begins. We want experts to have more warm up tosses than non-experts, so we can set the number of warm up tosses equal to their expertise level multiplied by 50. This gives agents a number of warm up tosses between 0 and 10 based on their expertise level.[8] The simulation then proceeds as normal according to the track records model described above. Let's call this the "expert priors" model.

I performed 10,000 runs of the expert priors model for 200 time steps and averaged the results across runs. The results are shown in Figure 3. We can see that the model initially performs better than the baseline because agents' priors are informed by additional information. However, we see that the expert priors model quickly begins to underperform the baseline. This is surprising since the experts are starting with better priors and so should have better initial predictions and so be at the top of the track records leaderboard. However, because the track records are based on sparse, limited data, non-experts can still get lucky early on and mislead the community as Huang as identified. This shows that Huang's findings are robust across at least two different kinds of expertise: interpretational expertise and prior expertise.

However, having strengthened Huang's case for the potential misleadingness of track records, I'll now identify three features of realistic academic communities which plausibly

[8] Recall that agents' expertise levels are uniformly-randomly drawn from [0,0.2]. Multiplying this by 50 gives a range of [0,10].

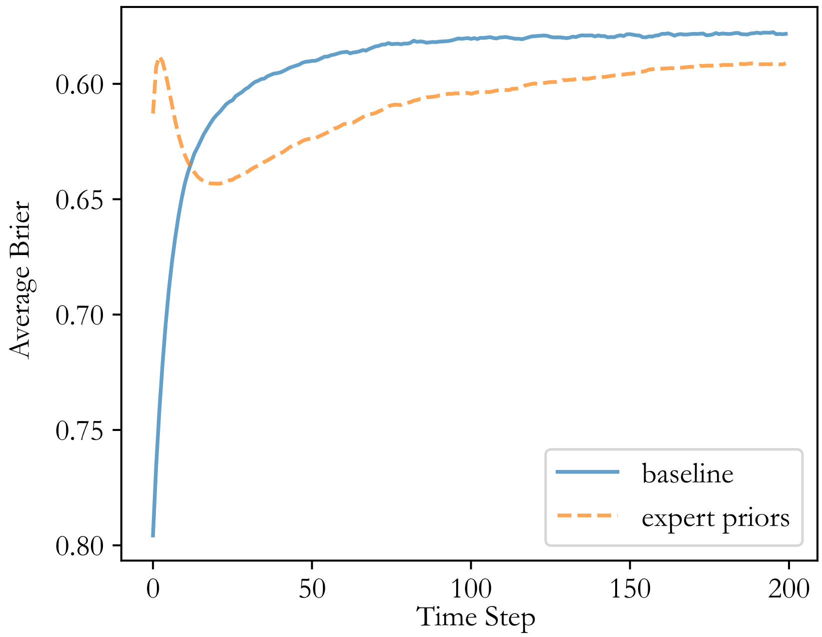

Figure 3: Expert Priors Model

make track records effective early on.

4 Unclear and Delayed Track Records

The second modification I want to consider has to do with the *clarity* of track records and *when* track records are obtained. In Huang's model, track records are perfectly clear and immediately available to all agents. It's precisely such features which Huang thinks makes her findings more significant, since they show that even under ideal conditions, track records can fail to identify experts. How much less useful should they be in the messiness of the real world! However, I'll show that these ideal conditions actually *disadvantage* track records and make them less effective.

Let's start with the issue of clarity. In Huang's model, track records are precisely numerically calculated Brier scores. Needless to say, the Brier scores of real-world experts aren't so precisely available. An important reason for this is because many predictions and/or outcomes are *qualitative* rather than quantitative and so it's not clear what exactly an agent's prediction was nor whether it came true or not. As a result, agents often differ as to what an agent's track record is and so as to who the experts really are.

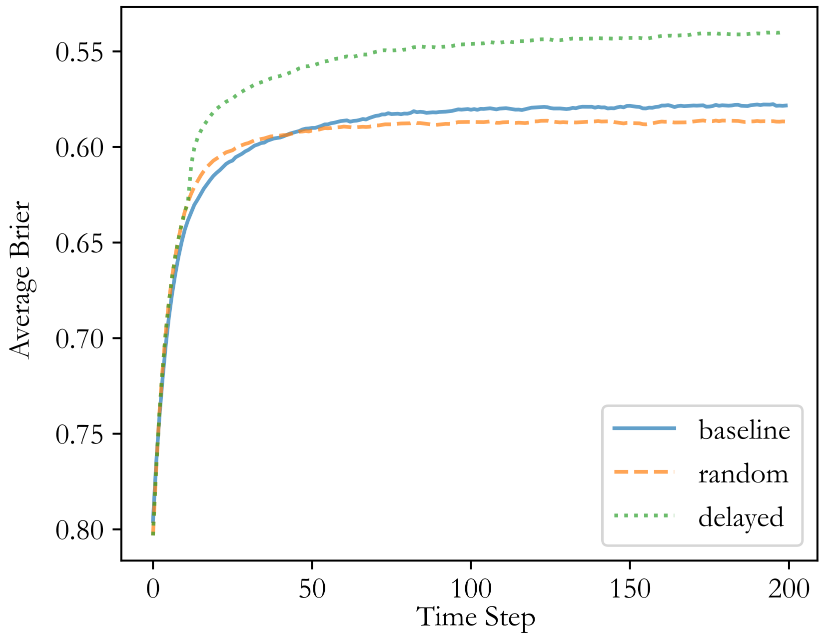

Figure 4: Delayed Track Records Model

One way to model this is to vary the proportion of agents who choose their informants randomly rather than based on track records. If track records were perfectly clear, then agents would behave as in Huang's original track records model. If track records are completely unclear, then agents would behave as though choosing their informants randomly. Note that Huang's less monopoly model falls in between these two extremes with half of agents choosing randomly. I'll consider the case where all agents choose their informants randomly. Let's call this the "random" model.

The results of the random model are shown in Figure 4. We can see that the random model actually outperforms the baseline model for the first forty or so time steps. However, after this point, the random model begins to underperform the baseline. This is the opposite behaviour of the less monopoly model which initially underperforms the baseline but then eventually outperforms it after around one hundred twenty time steps (see Figure 2).[9] Why does the random model perform better early on? Since agents choose their informants randomly, they avoid the problem of opinion monopoly which plagues

[9] This suggests that track record agents underperform random agents early in inquiry but outperform them later in inquiry. As a result, it seems that the track records model will eventually outperform the baseline model and the less monopoly model given enough time.

the track records model. Agents randomly consolidating their opinions with one another turns out to benefit the non-experts more than it hinders the experts early on and so increases overall accuracy. However, this indiscriminate belief-mixing also hinders the experts later so that the average performance is worse than the baseline.

This is a straightforward way to show that track records can be helpful early on if they're sufficiently unclear. Huang (2023, fn. 17) claims that her results were qualitatively similar with varying ratios of track record agents to random agents but this is mistaken, possibly because she wasn't performing enough simulation runs. Nevertheless, this isn't really a satisfying way of showing that track records can be helpful early on since (i) it's not particularly realistic to suppose that all agents would choose their informants randomly because of the collective action problem Huang identifies and (ii) the track records perform *worse* than the baseline in the long-run which doesn't make it advisable public policy.

However, by incorporating another realistic feature of communities we can address both of these issues. This feature has to do with *when* track records are obtained. In Huang's model, agents obtain track records immediately after each prediction. In real-world academic communities, however, data is typically only made public after some delay. The clearest way this occurs is through peer review, which forces researchers to get their data scrutinized by others before it's publicly endorsed. The peer review process typically takes anywhere from a couple of months to over a year, depending on the field and the journal/publisher. This create a strong lag between researchers making predictions and their track records being made public.

We can model this by having agents wait some number of time steps T before their track records are made public. On step $T + 1$, agents' track records from their first prediction are made public. There is thus a lag of T time steps between when an agent makes a prediction and when their cumulative track record incorporating that prediction is made public. Prior to this, we can assume that agents randomly choose their informants since they have no track record information to go on. We can call this the "delayed" model.

I simulated 10,000 runs of the delayed model for 200 time steps with a delay of $T = 10$ time steps. The results are shown alongside the baseline and random models in Figure 4. We can see that, as expected, the delayed model performs identically to the random model for the first ten time steps. At this point, agents start to receive track records and their performance significantly increases. This shows that track records can be helpful early on if there is a delay in obtaining them. The immediacy of the track records in Huang's model is actually an important part of why track records fail to identify experts early on.

Why does the delay in track records make them useful? The explanation isn't the same as for Huang's patience model since agents aren't obtaining track records based on *all* of agents' ten predictions but just the first prediction once they start receiving track records. Instead, the reason is because the delay allows for track records to be based on more *independent* predictions. During the first T steps, agents' predictions are based on their own updating and a random selection of informants. Since agents aren't all using the same informants, their predictions are more independent of one another and so a better measure of their expertise. The delay of T time steps allows for T time steps of these more-independent predictions to be accumulated and fed to agents during time steps $T + 1$ through $2T$. While this track record data will lead to opinion monopoly, as in the track records model, by the time agents start to receive it, any non-experts with lucky early track records will have had their opinions informed by T time steps of evidence and the opinions of others. This means that their personal probabilities will be much less harmful to the community than when they are used immediately as in the track records model. As the time step increases, agents get track records based on more predictions that were acquired from later on in the first T time steps. These track records will be a better gauge of the true experts and allow the experts to distinguish themselves from the non-experts. It's at this point where track records become truly helpful to the community since now agents are all listening to the true experts.

So we see that in the case where track records are delayed, they can outperform the baseline model from early on. This means that in real-life systems where track records are delayed, we should expect track records to be helpful at all times rather than just later in inquiry. Huang's conclusions as to the harms of track records only apply to cases where track records are immediately available. While this may seem to be helpful to track records, it's actually a harmful idealization which makes them seem less effective than they would actually be expected to be.

5 Social Inertia

Another source of issues in Huang's model is that agents rely too heavily upon track records too quickly. Due to the limited, noisy evidence which is available early on, track records are unreliable early in inquiry. This leads to the problem of opinion monopoly where early lucky agents are treated as experts and so mislead the entire community. Agents place just as much weight upon track records when they're based on one or two data points as when they're based on hundreds of data points.

However, in real-world communities, it's highly plausible that agents would be more cautious about relying upon track records when the sample sizes informing them are small. If I pick the super bowl winner correctly one year, I'd hardly be considered an expert in predicting future winners. You'd want to see something of a history of accurate predictions before you'd consider them an expert.[10]

This suggests a natural way in which opinion monopoly can be avoided: Don't give too much weight to track records when sample sizes are small. The patience model incorporates this insight by allowing agents to obtain a 50-sample track record before agents begin relying upon track records. However, as Huang correctly mentions, this model is unrealistic since it requires all agents to ignore track records for a sustained period of time. We can avoid this problem by allowing agents to *gradually* build trust in track records as sample sizes increase rather than requiring them to completely ignore track records entirely for a sustained period of time.

There's no doubt myriad ways to do this, but I'll present one model, which I'll call the "inertia" model because it requires agents to resist the influence of early track records. In this model, we modify an agent's social weighting factor c according to the following equation:

$$c^* = 1 - \left(\frac{n - 4S_{\text{best}}}{n + I}\right)(1 - c) \tag{2}$$

In this equation, n is the number of predictions that have been made, S_{best} is the best Brier score among all agents, and I is a free parameter representing the social inertia of the agent. The idea is that as n grows, the term $(n - 4S_{\text{best}})/(n + I)$ will gradually grow from near 0 to 1 and thus c^* will gradually change from near 1 to c. For a full motivation of this formula, see Appendix A.

I ran this model for 10,000 runs of 200 time steps with $I = 25$. The results are shown in Figure 5. As we can see, the inertial model performs as well or better than the baseline for all time steps. This shows that track records can plausibly be useful in the short-term if agents are appropriately cautious about relying on them when sample sizes are small.

Do we still run into a collective action problem? The only agents whose self-interest it is to not de-weight low sample size track records are non-experts who have gotten lucky. This would seem to be a low fraction of the total scientists in a realistic community and so there doesn't seem to be any conflict of interest between individual and group

[10] This connects with condition (iii) from Section 6.

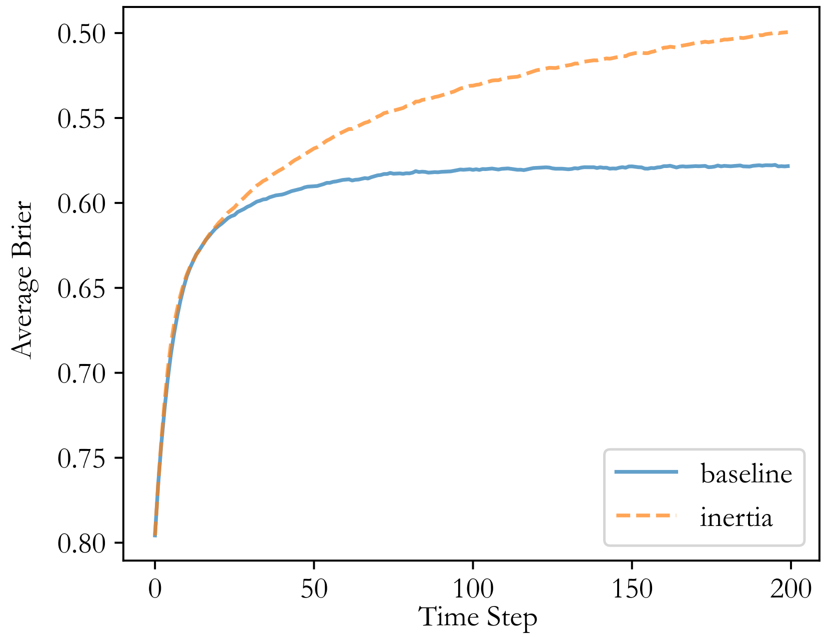

Figure 5: Inertia Model

epistemic goals for most agents. There thus doesn't seem to be any practical limitation in policymakers encouraging caution about track records in young fields.

An important limitation of this finding is that it depends on which c^* function one uses. For example, if one uses a linear growth function for c^* rather than the one I proposed, then track records don't perform as well in the short-term. This shows that the precise way in which agents build trust in track records matters.

While I applied the idea of gradual parameter growth to c, it also seems reasonable that one could also apply it to m, the openness parameter. That is, agents could gradually increase how many informants they consider as their trust in track records increases. I leave exploring this possibility to future work.

6 Public Evidence

A final modification I want to consider has to do with the public availability of evidence. Recall that in Huang's model evidence isn't shared. All agents have access to is other agents' track records (to choose their informants) and other agents' opinions (to calculate their social component). As Huang says, in her model agents unrealistically have perfect access to both of these. This would seem to be a problem for track records, since even if we have perfect access to them, they're either misleading or at best not helpful early in inquiry.

However, this aspect of the model is problematic for two reasons. First, in real-world academic communities, researchers typically have a common set of publically-accessible data. While they may have their own labs in which they and their graduate students perform many unsuccessful experiments which never see the light of day, their careers depend upon producing publishable data or interpretations thereof for the rest of the community.

Second, it doesn't make sense for agents to have track records but not the underlying evidence which produced those track records. For someone to believe that I'm an expert at picking the outcomes of some process X, they would want to see evidence (i) that I'd made predictions of outcomes before they occur, (ii) confirmation that the predictions came true, (iii) that my accuracy level is unexpected to occur by chance, and (iv) that the data hasn't selectively ommitted my failures to make me look better than I actually am.[11] For example, if I claim that I'm an expert at predicting the outcomes of football games, you'd want to see evidence of cases where I made predictions before the games occurred, that the teams that I predicted to win actually won, that my accuracy would be unexpected to occur by chance (one prediction won't cut it), and that my successful predictions haven't been cherry-picked . Element (ii) requires that the evidential outcomes be public. If all we have access to is someone making predictions for which I don't know whether they came true or not and I know that it's in their best interest to claim that they've made successful predictions, then, all else being equal, I won't trust their claims.[12] The issue isn't that this is an idealization, but that this is an idealization which is both unrealistic *and necessary* to generate the phenomena which Huang wants to highlight, as we'll see.

There are two distinct ways in which we could make the evidence public in Huang's model. First, we could suppose that agents share their individual evidence with one another. This would best represent situations where you have agents doing their own

[11] Every year before the Super Nowl there's news coverage of some animal along the lines that it has correctly predicted, say, the last five Super Bowl winners in a row. While this seems somewhat impressive because there's only about a 3% chance of randomly picking the correct winner five times in a row, it's not that impressive when you consider how many animals have no doubt been forced to make such predictions.

[12] Perhaps we could avoid this second problem by supposing that there is some neutral third-party who verifies agents' track records but doesn't share the underlying evidence. The problem is we don't live in such a world, and since such a world seems to lead to problems with track records, it doesn't seem desirable to bring about such a world.

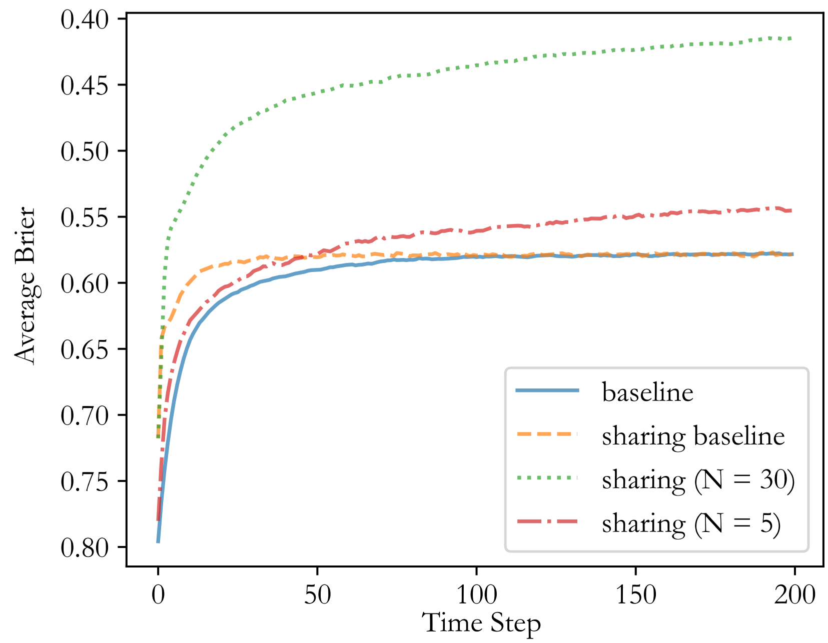

Figure 6: Evidence-Sharing Model

experiments and then publishing the results. Second, we could suppose that agents share the same one piece of evidence at each time step. This would best represent Big Science: Large-scale collaborative experiments involving many researchers. Examples would include the Human Genome Project (HGP), which involved around 2,800 researchers, or the Planck satellite, which involved around 600 researchers.

6.1 Shared Individual Evidence

Consider first the case where agents continue to get their own individual evidence but now share it with one another. We can call this the "sharing" model. The model is identical to the track records model except that now agents update their evidential components using not only their own data but also the data of everyone else at each time step. We can compare this to a new baseline where evidence is also shared but all agents don't give any weight to their social component ($c = 1$ for all agents). The results are shown in Figure 6.

Looking at the results, we can first compare the new baseline (with sharing) to the original baseline (no sharing). We can note that evidence-sharing alone keeps the final baseline accuracy the same but allows agents to reach it more quickly. This is because agents now get evidence much faster but the final accuracy level is still limited by the expertise levels of the agents.

Compared to this new baseline, we see that evidence-sharing allows track records to perform not just as well as the baseline but significantly better from the very start of inquiry. This finding can't be explained away as the result of agents simply having *more* evidence since the new baseline also includes evidence-sharing. Rather, evidence-sharing positively impacts the efficacy of track records. The reason for this is because agents now have significant data upon which to update their personal probabilties. This allows experts to distinguish themselves from non-experts much more quickly by having significantly more accurate personal probabilities and predictions.

In Huang's model, agents only have access to one bit of evidence per time step: The outcome of the coin flip. This is an essential part of why track records don't work well in her model: They're built upon extremely sparse, noisy data. One could argue that this is realistic since, in some cases, experiments are costly and so we only have access to limited data. The problem with this response is that, if experiments are costly, then it's unlikely that every agent could perform their own experiment. We'd be much more likely to have a Big Science case where many agents share the same data rather than each having their own private data. I'll treat this case in Section 6.2. I would just point out now that, in such scenarios, it would be even *more likely* that the data would be shared among agents rather than kept private. For instance, the datasets of the HGP and Planck satellite are publicly available for anyone to analyze.

A better response is that the assumption of sparse evidence is realistic in small research communities where few agents are working on a problem. In this case, there would be less evidence to go around. To check this possibility, I ran the evidence-sharing model with five agents rather than thirty to see if track records would still be effective. The results are in Figure 6. We can see that even with far fewer agents the sharing model outperforms the baseline after 40 time steps. This limits Huang's conclusion to small research communities. While such communities do exist, and are perhaps becoming more common through increased research specialization, they also don't seem to be the kinds of communities that are most of interest to public policymakers. Most of the important scientific expertise issues which face society—cancer, climate science, vaccines, AI safety, etc.—involve large research communities. Track records will be useful in such communities from the very start of inquiry. Huang's findings will only apply to groups much smaller than the size of communities she was studying.

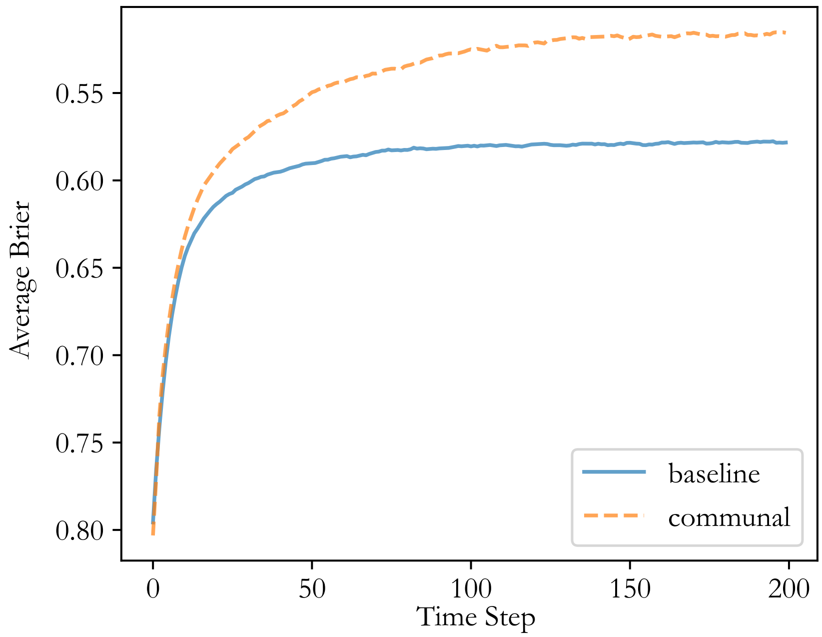

Figure 7: Communal Evidence Model

6.2 Communal Evidence

I mentioned earlier that another way to make evidence public is to have agents share the same piece of evidence at each time step. In this case, the agents don't share their individual evidence but possess the same communal evidence. This would best represent Big Science scenarios where many researchers collaborate on the same experiments. We can modify the track records model so that agents still only get one bit of evidence per time step but that it's the *same* bit of evidence. We can call this the "communal" model. The results of this model are shown in Figure 7.

In this case, we compare the communal model to the original baseline since both models involve agents getting only one bit of evidence per time step. We can see that communal evidence also allows track records to be useful in the short-term, though not as useful as in the sharing model. In this case, the advantage isn't due to the agents having more evidence or less noisy track records. Rather, the reason for the advantage is more subtle.

To understand why we can consider an analogy. Suppose there are good and bad students taking a test. 80% of the time a normal test is given where the good students pass and the bad students fail. 20% of the time a weird test is given where the good students fail and the bad students pass. If students are individually assigned tests according to these probabilities, then among the students who pass, there will always be some bad students who got lucky by being given the weird test. However, if all students are given the same test (according to these probabilities), then 80% of the time, all of the good students will pass and all of the bad students will fail and 20% of the time, all of the good students will fail and all of the bad students will pass. In the first case, there will *always* be *some* bad students who passed the test. In the second case, there will *rarely* be *many* bad students who passed the test.

The key is to see that second case provides greater average accuracy in identifying the good students than the first case. In each case, there is systematic luck for a given simulation run where the frequencies of exams given differs from the actual propensities. However, in the first case, there is also non-systematic, individual luck where some weird tests are given. This additional source of luck makes whether a student passed or failed less effective at identifying the good students on average.[13]

Likewise, in Huang's original model, since each agent gets their own private sequence of coin flips, we have thirty sequences rather than one. That makes it common for some non-experts to get lucky with a good initial track record from an anomalous sequence. This infects the top of the track records leaderboard with non-experts early in inquiry which leads other agents astray. In our new model, the agents share the same sequence of coin flips. This eliminates non-systematic data luck as a variable. While this will sometimes lead to an anomalous sequence where many non-experts get lucky, most of the time they won't. This allows track records to be more effective at identifying experts on average. So, the communal model gives us a second way in which public evidence can make track records effective from the start of inquiry.

7 Conclusion

Huang's conclusion is more powerful than she makes it out to be, for it holds for both interpretive and prior expertise. However, we've seen that there are at least three plausible ways in which Huang's negative results about track records can be avoided. First, if track records are *delayed* then agents have more independent track records which better track

[13] It's for this same reason that in modern portfolio theory it's better to diversify your investments to eliminate non-systematic risks due to specific industries or geographies, for example.

their actual expertise. Second, if agents are *cautious* about relying upon track records when sample sizes are small, then they won't give too much weight early on to those with the best track records and so won't give non-experts early opinion monopoly. Third, if agents' evidence is *public*, then this inhibits non-experts from getting lucky and allows experts to shine. Since real-world academic communities typically involve one or more of these three features, it's probable that track records are helpful from the very start of inquiry in real-world communities. Should we use track records to identify experts? Almost always, yes.

References

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A Social Inertia Model

The idea behind the inertia model is to let c start at 1 and then gradually lower it as agents acquire more data (and hence track records become more reliable). A simple way to do this is with the following equation:

$$c^* = 1 - T(1 - c) \tag{3}$$

The factor c^* is the effective social-weighting factor and T is a trust factor which represents how much an agent trusts the current track records (described below). If an agent has no trust in track records ($T = 0$), then their c^* value remains 1 and they just update their credences using their individual evidence. If an agent has full trust in track records ($T = 1$), then their c^* value becomes c, the weight they would naturally assign to track records.

So what should the trust factor T be? I propose, somewhat arbitrarily, to set it as follows:

$$T = 1 - \left(\frac{\bar{S^*}_{\text{best}}}{\bar{S}_{\text{chance}}}\right) = 1 - \left(\frac{\bar{S^*}_{\text{best}}}{0.25}\right) \tag{4}$$

$\bar{S^*}$ is a modified Brier score which adjusts for small sample sizes (described below). $\bar{S^*}_{\text{best}}$ is the best such score among all agents while $\bar{S}_{\text{chance}} = 0.25$ is the Brier score that an uninformed agent would get on average by random guessing.[14] Thus, Equation (4) sets an agent's trust in track records to be proportional to the improvement which the best agent has made over an uninformed guesser in their modified Brier score. If the best agent has made little improvement over an uninformed guesser, then agents will have little trust in track records, and vice versa.

So what is this modified Brier score $\bar{S^*}$? The track record of agents in Huang's original model is calculated using their total brier score S and the number of predictions they've made n. Their track record score is simply $\bar{S} = S/n$ which gives their average brier score. We want a function that will be close to $\bar{S}_{\text{chance}}$ when sample sizes are small (so that T is small) and will approach the actual average Brier score $\bar{S}$ as sample sizes get large (so that T is larger). Inspired by Laplace's urn solution, I propose the following modification:

$$\bar{S^*} = \frac{S + \bar{S}_{\text{chance}} I}{n + I} = \frac{S + 0.25I}{n + I} \tag{5}$$

[14] Note that the average Brier score of an uninformed guesser who estimates the probability of heads as $1/2$ on each toss will be 0.25.

Here, I is a inertial factor which represents how much weight we give to the default uninformed Brier score of 0.25 when sample sizes are small. Before we have any data ($S = n = 0$), an agent's track record score will be the uninformed score of 0.25. As the agent gets data, the S and n terms grow in significance relative to the inertial terms, and their modified Brier score will approximate S/n which is just their actual average Brier score $\bar{S}$.

We can combine Equations (3) to (5) to obtain the following expression for the modified social weighting factor:

$$c^* = 1 - \left(\frac{n - 4S_{\text{best}}}{n + I} \right) (1 - c) \tag{6}$$

Let's not get lost in the math at this point. The idea is simple: Let an agent's social reliance on track records build gradually as the sample sizes informing the track records increase.

Let's see a concrete example of how this works. Suppose an agent has an initial credence of 0.7 that the coin will land heads and it does land heads. Their average Brier score would be $\bar{S} = 0.09$. However, another agent with inertia $I = 10$ would assign them a track record score of $\bar{S}^* = [0.09 + 0.25(10)]/[1 + 10] \approx 0.2355$. This is much closer to the uninformed guesser score of 0.25. Suppose that the agent still has this average Brier score after 20 predictions. In that case, their track record score would be $\bar{S}^* = [1.8 + 0.25(10)]/[20 + 10] \approx 0.1167$. Their average Brier score hasn't changed but due to the larger sample size, their track record score is now lower. This will have the effect of increasing T in Equation (4) and so increasing c^* in Equation (3). Thus, our model allows for gradual trust in track records to build and so for trust in others' opinions to build.